

Differential Gene Expression Analysis & Molecular Network Reconstruction of RNA-Seq Data in an Asthma Disease Model (Human Chromosome 21)

Method: Linux WSL2, Bowtie2, StringTie, DESeq2, AnnotationDbi, & STRING DB v12.0

GitHub: <https://github.com/gracecieloisa/RNASeq-Asthma-DGE-Analysis>

ABSTRACT

Unraveling molecular mechanisms in inflammatory airway diseases such as asthma requires a transcriptomic approach. This study applies an end-to-end RNA-Seq pipeline on human Chromosome 21 data (GRCh38) to compare gene expression profiles between Asthma & Control groups. Upstream processing using Trimmomatic and Bowtie2 achieved a sequence alignment rate of 97.98%. Transcript quantification was performed with StringTie and prepDE.py. Downstream biostatistical analysis using DESeq2 combined with Regular Expression-based feature cleaning successfully reduced annotation missing values from 95.38% down to 6.92% (achieving an annotation success rate of 93.08%). The analysis identified 7 key Differentially Expressed Genes (DEGs) ($P_{adj} \leq 0.05$ and $|\log_2 FC| \geq 1.0$). Protein-Protein Interaction (PPI) network reconstruction on STRING DB revealed a dual pathophysiological axis: suppression of innate immunity / anti-inflammatory defense (drastic downregulation of IL10RB and MX1) and glycolytic metabolic reprogramming (massive upregulation of PFKL and SUMO3).

1. INTRODUCTION

Asthma is a chronic inflammatory airway disease involving complex interactions among genetic factors, metabolic stress, and immune responses. With the advent of RNA Sequencing (RNA-Seq) technology, Differential Gene Expression (DGE) analysis has become the gold standard method for mapping transcriptomic alterations at single-nucleotide resolution. This report presents a concise summary of the methodology and results from comparative transcriptomic data between Asthma and Control groups on human Chromosome 21, serving as a publication-grade validation model for bioinformatics pipelines.

2. MATERIALS AND METHODS

2.1 Upstream Pipeline & Sequence Processing

Raw sequencing reads were accessed in a Linux Ubuntu 22.04 LTS environment via Windows Subsystem for Linux (WSL2). Initial quality control was conducted using FastQC v0.11.9 to evaluate Phred score distributions and adapter content. Trimming of adapters and low-quality bases was executed using Trimmomatic v0.39 with parameters LEADING:20 TRAILING:20 SLIDINGWINDOW:4:20 MINLEN:50. Clean reads were aligned to the human Chromosome 21 reference genome (GRCh38) using Bowtie2 v2.4.5. Output SAM files were converted into sorted BAM files and indexed using Samtools v1.13.

2.2 Transcript Quantification & Count Matrix Generation (Midstream Pipeline)

Transcript abundance was estimated using StringTie v2.2.1 guided by Ensembl GTF reference annotation. The raw integer count matrix was extracted from StringTie quantification folders using the Python script prepDE.py to produce the gene_count_matrix.csv file as the mandatory input for statistical analysis in R.

2.3 Biostatistical Analysis & Gene Annotation Cleaning (Downstream Pipeline)

Differential expression analysis was executed in RStudio using the Bioconductor package DESeq2 v1.38.0. Normalization was performed via the Median-of-Ratios method to correct for differences in sequencing depth. Initial filtering was applied to remove genes with $\text{rowSums(counts)} < 10$. Variance Stabilizing Transformation (VST) was executed using `varianceStabilizingTransformation()` for dimensionality reduction via Principal Component Analysis (PCA). Gene symbol annotation was performed using AnnotationDbi and the org.Hs.eg.db database. To resolve missing values caused by StringTie headers (pipe |) and Ensembl version suffixes (dot .), precision feature cleaning was applied using the R regular expression: `sub("[\\.|\\.]*", "", ensemble_raw)`.

2.4 Protein-Protein Interaction Network Reconstruction (STRING DB)

The list of significant DEGs ($P_{adj} \leq 0.05$ and $|\log_2 FC| \geq 1.0$) was submitted to the STRING Database v12.0 platform (*Homo sapiens*) to reconstruct the Protein-Protein Interaction (PPI) network and evaluate functional modules.

3. RESULTS AND DISCUSSION

3.1 Sequence Processing Performance & Feature Annotation Cleaning

Trimmomatic trimming successfully retained 42,526 reads (64.19%) of high-quality data (Phred Score > 30). Bowtie2 alignment to Chromosome 21 yielded a 97.98% overall alignment rate, confirming highly adequate data quality. At the downstream annotation stage, regular expression-based ID cleaning reduced missing symbol values from 95.38% (124/130 genes) down to just 6.92% (9/130 genes), achieving an annotation success rate of 93.08%.

3.2 PCA Dimensionality Reduction & Differential Expression Profile

VST-based PCA plot demonstrated clear group separation between Asthma and Control groups, accounting for 84% of total explained variance (PC1: 50%, PC2: 34%). Statistical filtering identified 7 key significant DEGs on Chromosome 21, summarized in Table 1 below:

Official Symbol	Log2FC	P-Adjusted (Padj)	Status	Full Gene Name & Primary Biological Function
PFKL	+8.73	7.76×10^{-6}	Upregulated	Phosphofructokinase, liver type (Rate-limiting enzyme of glycolysis / metabolism)
SUMO3	+24.07	8.42×10^{-6}	Upregulated	Small ubiquitin like modifier 3 (Post-translational modification & stress response)
IL10RB	-11.85	1.69×10^{-11}	Downregulated	Interleukin 10 receptor subunit beta (Anti-inflammatory cytokine receptor subunit)
KCNJ15	-25.39	4.83×10^{-6}	Downregulated	Potassium inwardly rectifying channel J15 (Cellular ion homeostasis)
MX1	-8.69	1.90×10^{-2}	Downregulated	MX dynamin like GTPase 1 (Innate immune response & interferon pathway)
MCM3AP	-8.76	6.10×10^{-5}	Downregulated	MCM3 associated protein (Cell cycle regulation & mRNA transport)
RPS26P4	-2.08	1.34×10^{-7}	Downregulated	Ribosomal protein S26 pseudogene 4 (Ribosomal pseudogene component)

Table 1. List of 7 Significant DEGs on Chromosome 21 (Asthma vs. Control, $P_{adj} \leq 0.05$, $|\text{Log2FC}| \geq 1.0$).

3.3 Molecular Mechanism & Protein-Protein Interaction (PPI) Network Interpretation

Mapping on STRING DB extracted a core connected component: IL10RB – MX1 – PFKL – SUMO3. This finding uncovers two primary pathophysiological axes in asthma:

- 1. Immune Dysregulation Axis:** Severe downregulation of *IL10RB* (Log2FC = -11.85) and *MX1* (Log2FC = -8.69) indicates the loss of the IL-10/IL-22 anti-inflammatory 'brake' system and the suppression of cellular interferon response pathways, leading to uncontrolled airway inflammation.
- 2. Metabolic Reprogramming Axis:** Massive upregulation of *PFKL* (Log2FC = +8.73) and *SUMO3* (Log2FC = +24.07) shows that inflammatory cells undergo enhanced glycolytic flux to satisfy high energy demands, balanced by sumoylation processes to stabilize cellular transcription factors under inflammatory stress.

4. CONCLUSION

This transcriptomic study confirms the reliability of the end-to-end RNA-Seq pipeline from upstream to downstream stages. The identification of 7 key genes on Chromosome 21 demonstrates that asthma pathophysiology involves a combination of innate immune suppression and a glycolytic metabolic switch. The application of regular expression-based annotation cleaning significantly improves omics data annotation quality, yielding valid biological insights suitable for international publication standards.